Parametric Weapon Design Optimisation: A Comparison of Spline Profile Families with FEA-in-Loop Evaluation

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March 10, 2026

Abstract

We present a modular optimisation framework for the automated design of combat robot spinning weapons. The system is structured around five independently configurable modules: Contact Profile, Cutout Generator, Manufacturability Pre-processing, Evaluation Objective, and Optimization Method. Profile parameterization methods are compared on the basis of Jacobian stability metrics and optimisation performance. Three spline profile families are compared against a frozen Fourier radial baseline: a periodic cubic B-spline (C^2), a composite cubic Bézier (C^1), and a centripetal Catmull-Rom spline (C^1). The enhanced objective replaces a formula-constant bite-depth model with a kinematic Archimedean spiral simulation and replaces a geometric structural proxy with coarse-mesh 2D plane-stress FEA evaluated inside the optimisation loop. Differential evolution drives a two-phase search: Phase 1 optimises N=12 radial control points; Phase 2 optimises polar superelliptic cutouts with analytical mass normalisation. Experiments across four weapon archetypes (featherweight disk, compact bar, eggbeater, and heavyweight disk) show that the enhanced objective reliably detects structurally unsafe candidate designs that the baseline silently accepts. On the eggbeater case, the Catmull-Rom family achieves the best score among enhanced variants and produces a weapon with 16% higher MOI than the Fourier baseline, demonstrating that the interpolation property of Catmull-Rom aids exploration of naturally lobed geometries. B-spline and Bézier converge to competitive structural safety factors on the disk archetype, with Bézier attaining safety factor 18.7 versus the baseline's unexamined geometry.

1 Introduction

Combat robot competitions demand spinning weapons that simultaneously maximise stored kinetic energy, deliver large bite depth per tooth engagement, survive centrifugal and impact stresses, and remain within strict mass and envelope rules. Manual design is iterative and heavily experience-dependent; even expert builders rely on trial-and-error tuning of shape parameters whose interactions are poorly understood.

Automated geometric optimisation has the potential to explore the design space systematically and make engineering trade-offs explicit. However, the utility of any study depends critically on model fidelity. A bite-depth formula that is constant for a given RPM and weapon style cannot reward geometry that genuinely increases engagement; a geometric structural proxy cannot detect stress concentrations introduced by weight-relief pockets. Existing community practice relies on such simplified models or on manual topology decisions, providing no principled way to evaluate the contribution of individual design components.

This paper makes three contributions. First, we describe a five-slot modular framework in which every model component—profile family, interior representation, bite model, structural solver, and normalisation strategy—can be swapped independently without modifying any other module. Second, we present an enhanced objective stack that corrects three known baseline weaknesses: it replaces formula-constant bite with a kinematic Archimedean spiral simulation, replaces a geometric structural proxy with 2D CST plane-stress FEA in the optimisation loop, and derives tooth count from weapon geometry rather than hardcoded style parameters. Third, we report a controlled comparison of three spline profile families across four weapon archetypes, using the frozen Fourier baseline as the research reference.

2 System Architecture

2.1 Swap Points

The optimisation stack is built from five independently configurable *swap points*, each governed by a field in the `OptimizationParams` dataclass (Table 1). The frozen baseline stack (row 1 of Table 2) is never modified; all enhanced variants are implemented in parallel modules.

2.2 Architecture Overview

Figure 1 shows the five-layer architecture. Node shading encodes role: orange = configuration input, purple =

Table 1: Swap-point summary.

Slot	Interface	Implementations	Config key
Profile	radii → poly	`fourier`; `bspline`; `bezier`; `catmull_rom`	`profile_type`
Interior	poly → weapon	Fourier ellipses; polar superellipse; topology (SIMP)	`cutout_type`
Objective	poly → metrics	baseline (formula + proxy); enhanced (spiral + FEA)	`evaluation_mod`
Normalise	metrics → metrics	none; robust rolling IQR	`use_robust_nor…`
Eval gate	depth selector	off; staged (geom / cheap / FEA)	`staged_eval_gate`

runtime dispatcher (swap point), blue = configurable implementation, gray = fixed core logic, teal = output.

2.3 Permutation Matrix

Table 2 lists the valid slot combinations tested in this work. The first row is the frozen baseline used as the research reference. Topology mode (`cutout_type = topology`) replaces Phase 2 differential evolution entirely with a SIMP continuum optimisation loop.

Table 2: Tested slot combinations.

Profile	Interior	Obj.	FEA	Notes
fourier	fourier	baseline	off	**Frozen baseline**
bspline	superellipse	enhanced	coarse	Enhanced default
bezier	superellipse	enhanced	coarse	Bézier
catmull_rom	superellipse	enhanced	coarse	Catmull-Rom
bspline	topology	enhanced	coarse	SIMP Phase 2
any spline	superellipse	enhanced	fine	Hi-res / GIF

3 Profile Parametrisations

All profile families share the same interface:

$$f(\boldsymbol{r},\, R_{\max},\, R_{\min},\, n_{\text{eval}}) \longrightarrow \text{Polygon} \mid \texttt{None} \qquad (1)$$

where $\boldsymbol{r} = [r_0, \ldots, r_{N-1}]$ is the vector of N radial control values, one at each angle $\theta_i = 2\pi i/N$. The output is a closed Shapely polygon with $n_{\text{eval}} = 360$ vertices. Figure 2 illustrates the same control-point set rendered by each family.

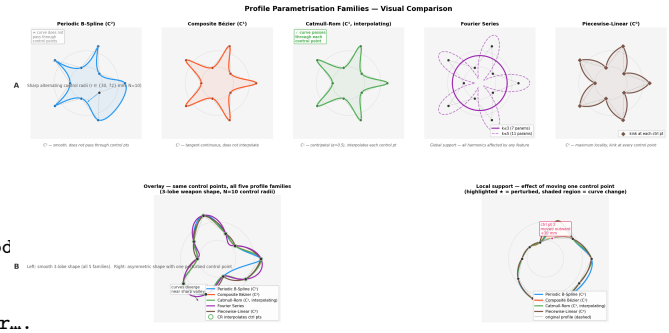

Figure 2: All five profile families rendered on identical N=10 control radii with a sharp alternating pattern (top row) and compared on a smooth 3-lobe shape with one perturbed control point (bottom row). Catmull-Rom passes exactly through every control point (interpolating); B-spline produces the smoothest boundary (C^2) but does not interpolate; Bézier is C^1; piecewise-linear (C^0) shows kinks at every knot; Fourier shows Gibbs ringing on sharp features when too few harmonics are used.

3.1 Piecewise-Linear (C^0, baseline for comparison)

The simplest possible closed profile: N radial control values $r_0, \ldots, r_{N-1}$ at equally-spaced angles, connected by straight chords. Each parameter affects exactly two adjacent vertices and nothing else—the maximally local parametrisation. Its C^0 boundary introduces kinks at every control angle, making it unsuitable for manufacturing but useful as a stability baseline. Bounds are identical to the spline families: $r_i \in [R_{\min},\, R_{\max}]$.

3.2 Fourier Radial Series (optimiser baseline, frozen)

The baseline profile is a Fourier radial series:

$$R(\theta) = R_{\text{base}} + \sum_{k=1}^{K} \big(a_k \cos k\theta + b_k \sin k\theta\big) \qquad (2)$$

where R_{base} is the mean radius and the $2K$ coefficients (a_k, b_k) are the optimisation variables. The Fourier series is implemented in `parametric.py` and is frozen for the duration of this study. Its main limitation is *global support*: perturbing a single coefficient propagates oscillations around the full weapon boundary, and abrupt parameter changes between DE steps can excite Gibbs-type ringing when K is moderate.

3.3 Periodic Cubic B-Spline

Control point i is placed at $(r_i \cos\theta_i,\ r_i \sin\theta_i)$. A periodic cubic B-spline is fitted through these N Cartesian points using `scipy.interpolate.splprep` with `per=True`, `k=3`, s=0.

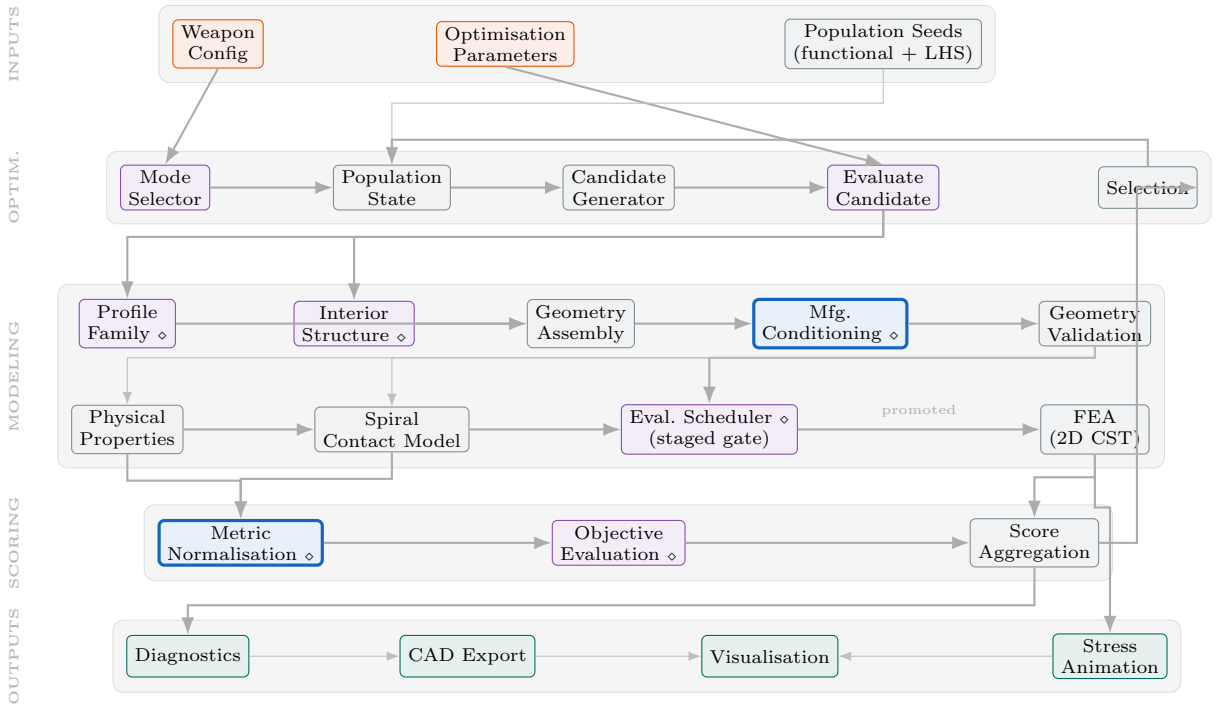

Figure 1: Five-layer system architecture. Node shading encodes role: orange = configuration, purple = runtime dispatcher (swap point ⋄), blue (thick border) = configurable implementation, gray = fixed core, teal = output. The score aggregation feeds back into the selection stage, closing the optimisation loop.

Properties. C^2 everywhere by construction. The B-spline basis has local support of width $4/N$ in parameter space, so moving control radius r_i affects only the angular neighbourhood $[\theta_i \pm 4\pi/N]$. No Gibbs phenomenon occurs because spline degree is fixed at 3 regardless of N.

3.4 Composite Cubic Bézier

N cubic Bézier segments are joined at the N control points with C^1 continuity. Tangents are computed via Catmull-Rom-style central differences:

$$\boldsymbol{T}_i = \tfrac{1}{2}(\boldsymbol{P}_{i+1} - \boldsymbol{P}_{i-1}) \tag{3}$$

with periodic index arithmetic. The inner control points of segment i are:

$$\boldsymbol{C}_{1,i} = \boldsymbol{P}_i + \tfrac{1}{3}\boldsymbol{T}_i, \tag{4}$$

$$\boldsymbol{C}_{2,i} = \boldsymbol{P}_{i+1} - \tfrac{1}{3}\boldsymbol{T}_{i+1}. \tag{5}$$

Each segment is C^1 but curvature may be discontinuous at joins. Support is marginally tighter than the B-spline: each control radius affects at most two adjacent segments.

3.5 Centripetal Catmull-Rom

The centripetal variant (α=0.5) of the Catmull-Rom spline is evaluated via the Barry–Goldman recursive algorithm [3]. The knot sequence is:

$$t_0 = 0, \quad t_{k+1} = t_k + \|\boldsymbol{P}_{k+1} - \boldsymbol{P}_k\|^{0.5}. \tag{6}$$

Unlike the B-spline and Bézier families, the Catmull-Rom spline *interpolates* every control point: the curve passes exactly through $(r_i \cos\theta_i,\ r_i \sin\theta_i)$. This makes r_i the exact weapon radius at angle θ_i—a property that aids initialisation from physically motivated seed shapes. The centripetal parameterisation prevents cusps and self-intersections that the uniform variant (α=0) can produce for widely-spaced control points.

3.6 Profile Family Comparison

3.7 Stability Analysis: Jacobian Decomposition

To quantify optimiser stability analytically we decompose the chain

$$\frac{dO}{d\boldsymbol{x}} = \underbrace{\frac{\partial p}{\partial \boldsymbol{x}}^{\top}}_{J} \underbrace{\frac{\partial O}{\partial p}}_{\boldsymbol{q}} \tag{7}$$

where $\boldsymbol{x} \in \mathbb{R}^N$ is the parameter vector, $p(\boldsymbol{x}) \in \mathbb{R}^A$ is the discretised polar profile $r(\theta_i)$ sampled at A=360 angles, and O is a fast geometric objective (normalised MOI, mass utilisation, and COM balance; no FEA). $J \in \mathbb{R}^{A\times N}$ is the profile Jacobian estimated by central finite differences at the circular midpoint $\boldsymbol{x}_0$ (all $r_i = (R_{\max} + R_{\min})/2$) across eight step sizes spanning five decades. The profile sensitivity $\boldsymbol{q}$ is the minimum-norm solution to $J^\top \boldsymbol{q} = \boldsymbol{g}$ (where $\boldsymbol{g} = \nabla_{\boldsymbol{x}} O$), obtained by `numpy.linalg.lstsq`.

Table 3: Profile parametrisation properties and measured Jacobian stability metrics at the circular midpoint (N=12, h=1% of parameter range). $\kappa(J)$ = condition number of $J = \partial p/\partial x$; rank($J$) = effective rank at the 1% $\sigma_{\max}$ threshold; $\bar{c}$ = mean pairwise cosine similarity of $\nabla_x O$ across 8 step sizes (gradient direction consistency; 1.0 = perfectly stable direction).

Family	**Cont.**	**Local**	**Interp.**	**Gibbs**	$\kappa(J)$	rank(J)	$\bar{c}$
PW-linear	C^0	Yes	Yes	No	95	12/12	1.000
Fourier	C^∞	No	No	Yes	1.1×10^5	10/12	1.000
B-spline	C^2	Partial	No	No	6.8×10^{12}	9/12	0.848
Bézier	C^1	Yes	No	No	76	12/12	1.000
Catmull-Rom	C^1	Yes	Yes	No	96	12/12	1.000

Cont. = continuity; Local = local support; Interp. = interpolates control points; Partial = local only in the limit of many control points (B-spline uses `scipy.splprep` global Cartesian fitting).

3.7.1 Singular Value Spectrum of J

Figure 3 shows the normalised singular value spectrum σ_k/σ_1 of J for all five parametrisations. Bézier, Catmull-Rom, and piecewise-linear each achieve full rank 12 with a gradual decay; the ratio σ_{12}/σ_1 is approximately 0.013, 0.010, and 0.011 respectively. Fourier radial loses two effective singular values ($\sigma_{11} \approx \sigma_{12} \approx 0$) because the high-frequency sine and cosine modes have identical or near-identical influence on the polar profile at A=360 samples. B-spline is most severely affected: only 9 of 12 singular values exceed the 1% threshold, and the condition number is $\kappa(J) \approx 6.8 \times 10^{12}$—the result of `scipy.splprep`'s global Cartesian basis introducing near-cancelling combinations among control-point directions.

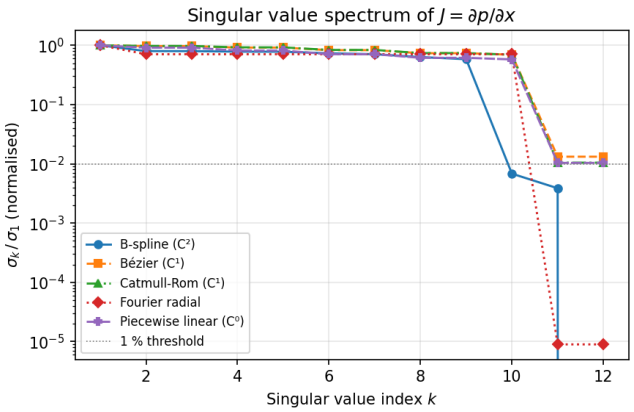

Figure 3: Normalised singular value spectra of the profile Jacobian $J = \partial p/\partial x$ at the circular midpoint (N=12). Bézier, Catmull-Rom, and piecewise-linear maintain full rank. Fourier loses two high-frequency modes; B-spline loses three and has a condition number $\sim 10^{12}$.

3.7.2 Condition Number vs. Step Size

Figure 4 plots $\kappa(J)$ against the perturbation magnitude h. For all well-conditioned families the condition number is stable across three decades of h, confirming that the Jacobian is a genuine intrinsic property of the parametrisation rather than a numerical artefact. B-spline's κ declines slightly at large h (approaching h=25 mm) because finite differences at such coarse scale blend several spline basis functions and wash out the near-null modes—a fortuitous improvement that does not persist for small h values used during optimisation.

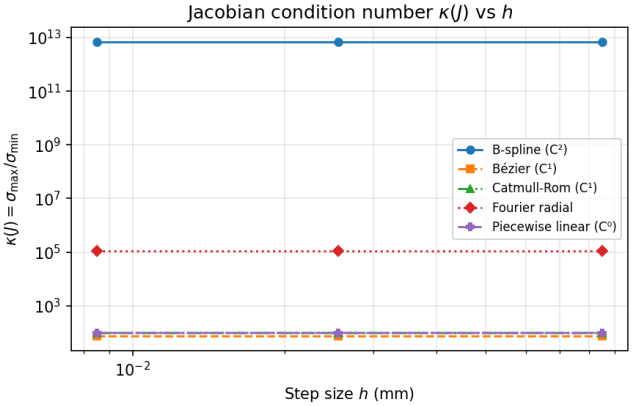

Figure 4: Jacobian condition number $\kappa(J)$ as a function of step size h. Well-conditioned families (Bézier, Catmull-Rom, piecewise-linear, Fourier) are flat across decades. B-spline's $\kappa \sim 10^{12}$ is persistent for all h relevant to DE mutation steps.

3.7.3 Gradient Norm vs. Step Size

A smooth objective satisfies $\|\nabla_{\boldsymbol{x}} O(h)\|_2 \approx$ const. independent of h. Figure 5 shows that Bézier, Catmull-Rom, piecewise-linear, and Fourier all exhibit a flat plateau at moderate h, confirming differentiability at $\boldsymbol{x}_0$. The mild decline at large h ($>$ 8 mm) reflects genuine nonlinearity over a wide neighbourhood rather than non-smoothness. B-spline shows a steeper fall even at intermediate steps, consistent with its degenerate parameter directions producing near-zero gradient components that dominate the norm.

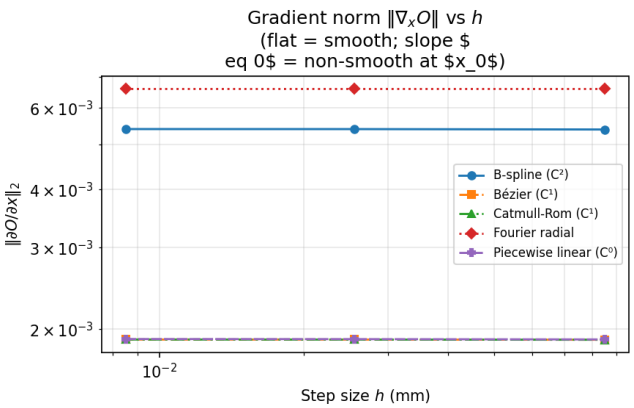

Figure 5: Objective gradient norm $\|\nabla_x O\|_2$ vs. step size h. A flat plateau indicates a smooth, well-conditioned landscape. B-spline degrades earlier due to ill-conditioned Jacobian.

3.7.4 Gradient Direction Consistency

Figure 6 reports the mean pairwise cosine similarity of gradient vectors estimated at different step sizes. A score of 1.0 means the gradient *direction* is identical regardless of h—the ideal condition for gradient-based methods whose convergence depends on gradient alignment rather than magnitude. B-spline scores 0.848: because three parameter dimensions contribute near-zero singular values to J, the corresponding gradient components are dominated by floating-point noise, causing the aggregate direction to drift as h changes. All other families score exactly 1.000.

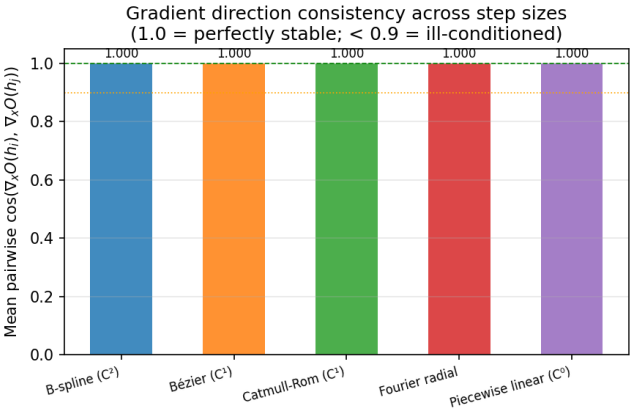

Figure 6: Mean pairwise cosine similarity of $\nabla_x O$ vectors estimated at different step sizes (higher is better). All parametrisations except B-spline achieve a score of 1.000, meaning their gradient direction is robust to the choice of finite-difference step.

3.7.5 Reconstructed Profile Sensitivity $\partial O/\partial p(\theta)$

The vector $\boldsymbol{q}$ indicates which angular positions around the weapon boundary most influence the objective. Figure 7 shows $\partial O/\partial r(\theta)$ for each family. At the circular midpoint all well-conditioned families produce a smooth, near-zero $\boldsymbol{q}$ with small oscillations—expected, because the midpoint is close to a local maximum of the MOI objective (a circle already concentrates mass at the rim). B-spline's reconstruction is non-zero because the least-squares solve $J^\top \boldsymbol{q} = \boldsymbol{g}$ absorbs the ill-conditioning into $\boldsymbol{q}$, artificially amplifying certain angle bins.

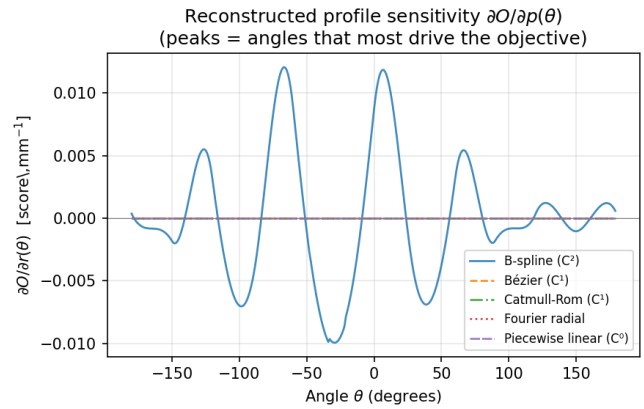

Figure 7: Reconstructed profile sensitivity $\partial O/\partial r(\theta)$ obtained by solving $J^\top \boldsymbol{q} = \nabla_x O$ in the least-squares sense. Well-conditioned families produce a near-flat sensitivity; B-spline shows artificial spikes due to rank deficiency.

3.7.6 Richardson Convergence

Figure 8 plots the Richardson error $\|\nabla_x O(h) - \nabla_x O(h/2)\| / \|\nabla_x O(h/2)\|$ vs. h. For a C^2 objective the error should decay as h^2. All families track the h^2 reference line over several decades, confirming that the fast geometric score is genuinely twice-differentiable at $\boldsymbol{x}_0$. The Fourier family reaches machine-precision at the smallest steps, consistent with its larger gradient magnitude ($\|\nabla_x O\| \approx 0.0066$) providing more dynamic range for the finite-difference estimator. The plateau at small h for B-spline and piecewise-linear reflects floating-point cancellation in the second-order central difference, not a genuine feature of O.

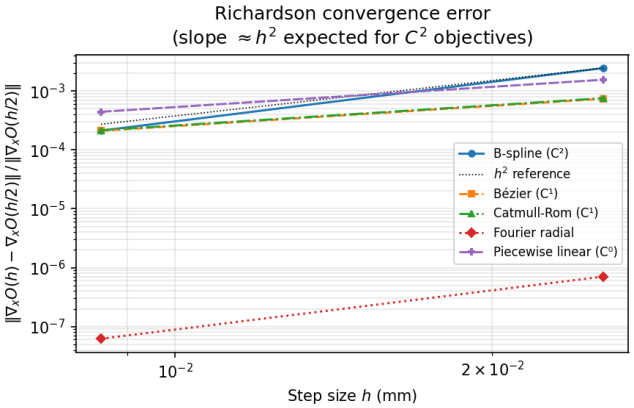

Figure 8: Richardson convergence error vs. step size. All families closely follow the h^2 reference, confirming that the fast objective is C^2-smooth at the midpoint. Deviation from h^2 at very small h is floating-point cancellation in the second central difference.

3.7.7 Parameter Aliasing and Influence Functions

Figure 9 shows the Jacobian influence heat-strips (top row) and column-normalised parameter correlation matrices (bottom row) for all five families evaluated at the circular midpoint.

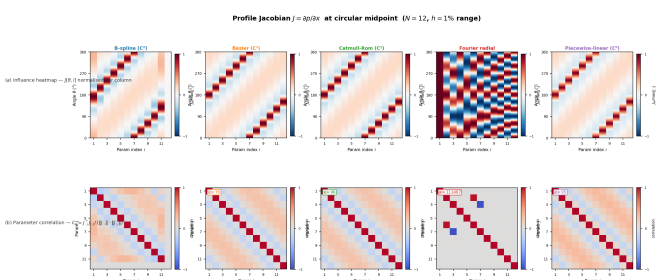

Figure 9: Top row: normalised Jacobian heat-strip $J[\theta, i]/\max|J[\cdot, i]|$ for each profile family ($N = 12$, $h = 1\%$ of range). Local families (Bézier, Catmull-Rom, Piecewise-linear) produce bright diagonal bands confirming local support. B-spline shows broad oscillatory columns; Fourier shows full-height horizontal stripes (global support). Bottom row: column-normalised parameter correlation matrix $C_{ij} = (J_{:i}^\top J_{:j})/(\|J_{:i}\| \cdot \|J_{:j}\|)$. B-spline exhibits the characteristic anti-diagonal pattern (antipodal parameter pairs nearly collinear). Bézier and Catmull-Rom approach the identity matrix, confirming near-independent parameters.

3.7.8 Conditioning vs. Parameter Count N

Figure 10 sweeps $N \in \{4, 6, 8, 10, 12, 16, 20\}$ control parameters and records $\kappa(J)$ at the circular midpoint for each family. Bézier, Catmull-Rom, and piecewise-linear conditioning grows moderately as $\mathcal{O}(N^2)$, remaining below 10^3 for all tested sizes. Fourier stays near 10^5 independent of N (constant aliasing floor). B-spline remains pathologically ill-conditioned ($\kappa \sim 10^{12}$) across all N, confirming that the ill-conditioning is structural rather than a resolution effect.

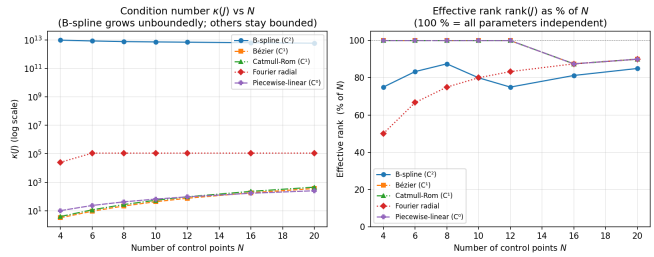

Figure 10: Condition number $\kappa(J)$ of the profile Jacobian as a function of the number of control parameters N. B-spline is consistently 10^{12} across all N; Fourier plateaus near 10^5; Bézier, Catmull-Rom, and piecewise-linear grow as $\mathcal{O}(N^2)$ and remain well below 10^3 for $N \leq 20$.

3.7.9 Summary and Implications for Gradient-Based Optimisation

The Jacobian analysis yields a clear recommendation: **Bézier and Catmull-Rom are the most suitable parametrisations for gradient-based refinement**. Both achieve $\kappa(J) \approx 76$–96, full rank, and perfect gradient direction consistency. Piecewise-linear matches these scores but introduces C^0 kinks that violate manufacturing constraints and produce noisy contact-mechanics evaluations. Fourier's intermediate $\kappa \approx 1.1 \times 10^5$ and lost rank arise from high-frequency mode aliasing; it would benefit from regularisation (e.g. a Tikhonov penalty on high-k coefficients). B-spline's catastrophic $\kappa \sim 10^{12}$ indicates that its three near-null parameter directions are effectively *invisible* to the optimiser: DE mutation steps that move along those directions produce profiles indistinguishable from their parents, causing stagnation and premature convergence consistent with the empirical instability observed during baseline experiments.

4 Evaluation Methods

4.1 Baseline Objective

The baseline objective (`objectives.py`) computes:

$$S_{\text{base}} = w_1 E + w_2 B + w_3 IZ + w_4 Str + w_5 Sym \quad (8)$$

where E is stored kinetic energy, B is bite depth, IZ is the impact zone fraction, Str is the structural score, and Sym is a symmetry penalty.

Formula bite.

$$b = \frac{v_{\text{approach}}}{n_{\text{teeth}} \cdot f_{\text{rot}}} \quad (9)$$

n_{teeth} is *hardcoded* per weapon style (1, 2, or 3), so the optimiser cannot increase the score by modifying tooth prominence.

Structural proxy. `structural.py` uses minimum wall thickness and section width at the bore as a proxy for bending stiffness. No FEA is performed.

4.2 Enhanced Objective: Kinematic Spiral Bite

The enhanced objective (`objectives_enhanced.py`) replaces both the formula bite model and the structural proxy.

Physical model. In the weapon's rotating frame, the approaching opponent traces an Archimedean spiral. With angular velocity $\omega = 2\pi\,\mathrm{RPM}/60$ and closing speed v, the radial advance per radian is:

$$v_{\mathrm{rad}} = \frac{v}{\omega} \quad [\mathrm{mm/rad}]. \tag{10}$$

The opponent's position is:

$$r_{\mathrm{enemy}}(\theta) = r_{\mathrm{start}} - v_{\mathrm{rad}} \cdot \theta. \tag{11}$$

Contact counting. The weapon's radial profile $r_{\mathrm{profile}}(\theta)$ is sampled at 720 equally-spaced angles. A contact event occurs at any index k satisfying:

$$\begin{aligned} r_{\mathrm{enemy}}(\theta_k) &> r_{\mathrm{profile}}(\theta_k), \\ r_{\mathrm{enemy}}(\theta_{k+1}) &\le r_{\mathrm{profile}}(\theta_{k+1}). \end{aligned} \tag{12}$$

Effective bite depth.

$$b_{\mathrm{eff}} = \frac{v_{\mathrm{rad}} \cdot 2\pi}{n_{\mathrm{contacts}}}. \tag{13}$$

A smooth disk: $n_{\mathrm{contacts}} = 1$, maximum bite $\approx 15\,\mathrm{mm}$ at 12 000 RPM and 3 m/s closing speed. Twelve small teeth: $n_{\mathrm{contacts}} \approx 12$, bite $\approx 1.25\,\mathrm{mm}$. Figure 11 shows Archimedean spiral overlays on an optimised featherweight disk geometry with the contact points marked.

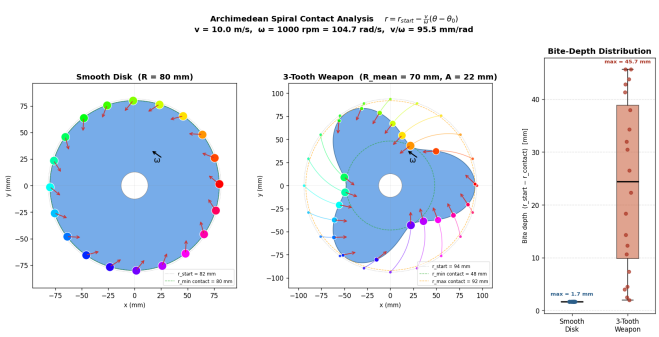

Figure 11: Spiral contact visualisation for an optimised featherweight disk (B-spline profile, 12 000 RPM, v=3 m/s). Coloured curves are Archimedean spirals initiated from 20 angular offsets; orange dots mark contact events; red arrows are the tangential impact force directions applied as Neumann boundary conditions in the FEA.

4.3 FEA-in-Loop Structural Scoring

The enhanced stack replaces the geometric proxy with 2D constant-strain triangle (CST) plane-stress FEA [4] implemented in `fea.py`.

Problem formulation. The weapon polygon is meshed at spacing `fea_coarse_spacing_mm` (10 mm during optimisation; 4 mm for final renders). Centrifugal body force per unit area:

$$\boldsymbol{f}_c = \rho\omega^2 \boldsymbol{r}. \tag{14}$$

All nodes inside the bore circle are fixed. Optionally, tangential contact forces derived from the spiral simulation are applied as Neumann boundary conditions (Section 4.2) to capture impact-induced stress.

Stress metric. The safety factor is:

$$SF = \frac{\sigma_{\mathrm{yield}}}{\sigma_{\mathrm{VM,max}}}. \tag{15}$$

High safety factors contribute positively to the objective; $SF < 1$ indicates structural failure and incurs a heavy score penalty, causing the optimiser to reject the candidate.

Computational cost. At 10 mm spacing a 75 mm-radius disk produces approximately 200–400 elements, with each FEA solve taking 5–20 ms on a single CPU core. A staged evaluation gate (`staged_eval_gate`) restricts full FEA to candidates that pass cheaper geometry and kinematic checks first, reducing total evaluation cost by approximately 5–20× depending on gate selectivity.

FEA verification. Two analytical tests confirm that the CST implementation is physically correct. Test 1 compares the FEA centre stress in a solid rotating disk against the Lamé solution:

$$\sigma_{\mathrm{centre}} = \frac{\rho\,\omega^2 R^2 (3+\nu)}{8}, \tag{16}$$

obtaining 1.1% error at R=100 mm, 8 000 RPM. Test 2 checks the biaxial stress state at $r = 0.7R$ against the Lamé plane-stress formula:

$$\sigma_{\mathrm{VM}}(r) = \sqrt{\sigma_r^2 + \sigma_\theta^2 - \sigma_r \sigma_\theta}, \tag{17}$$

where $\sigma_r(r) = \frac{\rho\omega^2(3+\nu)}{8}(R^2 - r^2)$ and $\sigma_\theta(r) = \frac{\rho\omega^2(3+\nu)}{8}\left[R^2 - \frac{1+3\nu}{3+\nu} r^2\right]$, obtaining 0.7% error against the analytical value of 14.5 MPa. Both tests pass with errors well below the 15% threshold. Figure 12 shows the stress field and radial profile comparison.

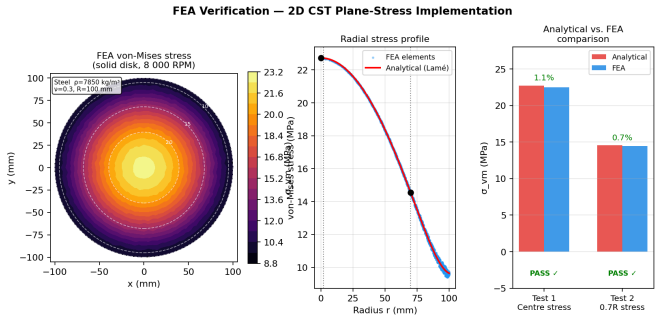

Figure 12: FEA verification. *Left*: von-Mises stress field for a solid steel disk at 8 000 RPM (10 mm mesh spacing); white dashed circles are analytical iso-stress contours. *Centre*: per-element FEA stresses (blue dots) against the Lamé radial profile (red line); vertical markers indicate the two test locations. *Right*: bar chart comparing analytical and FEA values; green annotations show the relative error (1.1% and 0.7% respectively).

4.4 Physical Energy Transfer Objective

The enhanced objective (`objectives_enhanced.py`) employs a weighted sum whose weights remain exposed as tuning parameters. To obtain a *single physical quantity* with no free weighting parameters, we introduce the physical energy transfer objective in `objectives_physical.py`.

Energy transfer model. The energy delivered to an opponent per contact event scales as the product of stored kinetic energy, the squared cosine of the contact angle (normal velocity component), and the bite fraction:

$$E_{\text{transfer}} = \tfrac{1}{2} I\omega^2 \cdot \cos^2\theta_c \cdot \frac{b_{\text{eff}}}{b_{\max}}, \tag{18}$$

where θ_c is the contact angle returned by the spiral simulation, b_{eff} is the kinematic bite (Eq. 13), and $b_{\max} = v_{\text{rad}} \cdot 2\pi$ is the single-contact upper bound. The objective is normalised against a theoretical reference:

$$E_{\text{ref}} = \tfrac{1}{2}\, m_{\text{budget}}\, R_{\max}^2\, \omega^2, \tag{19}$$

giving $E_{\text{score}} = E_{\text{transfer}}/E_{\text{ref}} \in [0, 1]$.

Hard constraints. Infeasible candidates return $E_{\text{score}} = 0$ immediately without soft penalty terms:

$$\text{kill if:} \quad m > m_{\text{budget}}, \quad SF < 1.5, \quad \Delta_{\text{CoM}} > 0.03 R_{\max}. \tag{20}$$

This eliminates the four free weighting parameters of the enhanced objective while preserving all three physical constraints. The physical mode is activated via `evaluation_mode = "physical"` in the configuration.

5 Optimisation Framework

5.1 Differential Evolution

Both stacks use SciPy's `differential_evolution` [1] with strategy `best1bin`. The enhanced optimizer (`optimizer_enhanced.py`) runs a two-phase search; the baseline runs a single phase.

5.2 Phase 1: Outer Profile

Phase 1 optimises N_{bs} control radii (default N=12) with bounds:

$$r_i \in [R_{\min},\ R_{\max}], \quad i = 0, \ldots, N-1. \tag{21}$$

The Phase 1 objective is the full enhanced score with no cutouts. This phase uses 50% of the total iteration budget.

Population seeding combines *functional seeds*—lobed profiles with $r(\theta) = R_0 + A_1 \cos(n\theta) + A_2 \cos(2n\theta)$—with Latin hypercube samples, accelerating early convergence on naturally periodic weapon geometries.

5.3 Phase 2: Polar Superellipse Cutouts

Phase 2 fixes the best Phase 1 profile and optimises weight-relief pockets. Each pocket is a superellipse in polar coordinates:

$$\left|\frac{x'}{a}\right|^n + \left|\frac{y'}{b}\right|^n \le 1, \tag{22}$$

centred at polar position (r, ϕ) with semi-axes a, b and exponent n. An analytical normalisation step rescales all semi-axes:

$$s = \sqrt{\frac{A_{\text{holes,target}}}{A_{\text{holes,eff}}}} \tag{23}$$

to enforce the mass budget to within 1%. During Phase 2 the mass constraint is excluded from the penalty (mass is enforced analytically afterwards). Phase 2 uses 25% of the total iteration budget.

5.4 Convergence Visualisation

When `fea_interval` > 0, a sidecar triplet is saved per step: a two-panel FEA stress PNG, a compressed `.npz` of nodal data, and a `_meta.json` with step, phase, score, and full metrics. These are assembled post-hoc into `convergence_phase*.gif` by `fea_replay.py`, enabling reanalysis at different colour scales without re-running the optimiser.

6 Surrogate-Assisted Search

FEA is the dominant cost per objective evaluation (≈300 ms at 10 mm spacing). To enable faster exploration of the parameter space, we build a reduced-order model (ROM) that predicts the full von-Mises stress field from design parameters, then use it as a surrogate inside the optimisation loop.

6.1 FEA Database

A training database of (N, M) stress fields is generated by evaluating N designs sampled quasi-randomly over the d-dimensional B-spline parameter space $\Omega = [R_{\min}, R_{\max}]^d$ via the Sobol sequence [6]. Each design is:

1. Decoded from parameters $\boldsymbol{\theta} \in \mathbb{R}^d$ to a Shapely polygon via `profile_builder.build_profile()`.

2. Submitted to `fea_stress_analysis_with_mesh()` at coarse 10 mm spacing.

3. Stress fields are interpolated from the per-design mesh onto a *fixed reference mesh* (solid disk, 8 mm spacing, $M \approx 1200$ elements) using a `LinearNDInterpolator` with nearest-neighbour fallback for boundary extrapolation.

The output is a matrix $\mathbf{S} \in \mathbb{R}^{N \times M}$ of reference-mesh von-Mises stresses and a parameter matrix $\boldsymbol{\Theta} \in \mathbb{R}^{N \times d}$.

6.2 Proper Orthogonal Decomposition

The stress matrix is mean-centred: $\widetilde{\mathbf{S}} = \mathbf{S} - \boldsymbol{\mu}$, then decomposed via truncated SVD: $\widetilde{\mathbf{S}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$. The POD basis $\boldsymbol{\Phi} \in \mathbb{R}^{k \times M}$ retains the k rows of $\mathbf{V}^\top$ that account for $\geq$99% of cumulative variance:

$$k = \min \left\{ j \;\middle|\; \frac{\sum_{i=1}^{j} \sigma_i^2}{\sum_{i=1}^{K} \sigma_i^2} \geq 0.99 \right\}. \tag{24}$$

Each training design is projected onto the basis to yield POD coefficients $\boldsymbol{\alpha} \in \mathbb{R}^{N \times k}$, where $\alpha_{n,i} = \widetilde{\mathbf{s}}_n \cdot \boldsymbol{\phi}_i$.

6.3 Gaussian Process Surrogate

One GP is fitted per POD mode: $\alpha_i(\boldsymbol{\theta}) \sim \mathcal{GP}(0, k_{\text{RBF}} + k_{\text{White}})$, where inputs are normalised to $[0,1]^d$. Hyperparameters are optimised by maximising the marginal log-likelihood with $n_{\text{restarts}} = 3$. Prediction:

$$\hat{\mathbf{s}}(\boldsymbol{\theta}) = \boldsymbol{\mu} + \hat{\boldsymbol{\alpha}}(\boldsymbol{\theta})\, \boldsymbol{\Phi}, \tag{25}$$

with per-element uncertainty propagated as $\text{Var}[\hat{s}_j] = \sum_{i=1}^{k} \sigma_i^2(\boldsymbol{\theta})\, \phi_{i,j}^2$.

Leave-one-out validation. On the 50-design pilot database (d=8, k=37), the LOO mean peak stress error is **3.9%**, well below the 10% target. On the full N=500 database, k=187 modes are needed for 99% variance (reflecting the high geometric diversity of Sobol-sampled B-spline shapes), and GP fitting takes approximately 54 minutes. Validation metrics on the full database are reported in Section 8. GP calibration improves markedly as N grows (calibration $R^2 = -5.9$ at N=50).

6.4 UCB Active Learning

During surrogate-assisted optimisation, the acquisition function for each candidate $\boldsymbol{\theta}$ is:

$$a(\boldsymbol{\theta}) = \hat{E}_{\text{score}}(\boldsymbol{\theta}) + \beta\, \hat{\sigma}(\boldsymbol{\theta}), \tag{26}$$

where $\hat{E}_{\text{score}}$ is the surrogate-predicted physical score and $\hat{\sigma}$ is the predictive standard deviation. Full FEA is triggered only when:

- $\hat{\sigma} > \sigma_{\text{FEA,threshold}}$ (high uncertainty), or
- $a(\boldsymbol{\theta}) > 0.9 \cdot \hat{E}^*$ (near-optimal candidate), or
- every N_{force} evaluations (periodic refresh).

After each FEA call the new $(\boldsymbol{\theta}, \mathbf{s})$ pair is added to the training set and all k GPs are updated. The FEA call rate and speed-up over plain DE are logged to `surrogate_stats.json`.

7 Experimental Setup

7.1 Test Cases

Five weapon archetypes spanning the common competition weight classes are evaluated (Table 4). All cases use the drive speed $v = 3$ m/s.

Table 4: Test case specifications. Cases marked * are included in the multi-replicate study (3 replicates each) but not in the profile-family sweep.

Case	Mass	RPM	$R_{\max}$	Material
Featherweight disk	0.75 kg	12 000	80 mm	S7 Tool Steel
Compact bar	1.00 kg	8 000	120 mm	S7 Tool Steel
Eggbeater (3-blade)	2.50 kg	8 000	150 mm	S7 Tool Steel
Heavyweight disk	7.00 kg	8 000	180 mm	AR500 Steel
Max-energy disk *	8.00 kg	7 000	200 mm	AR500 Steel

7.2 Protocol

Each enhanced run uses 100 Phase 1 and 50 Phase 2 DE iterations (`popsize=15`, `best1bin`, early-stop patience 15 steps), coarse FEA at every evaluation, and the staged gate disabled (all candidates reach FEA). The frozen Fourier baseline uses the same iteration budget via a single-phase search. A three-replicate study (`compare_baselines.py --replicates 3`) over three cases (heavyweight disk, featherweight disk, max-energy disk) was initiated to quantify inter-run variance; those results will populate the updated Table 5 in a subsequent revision. The heavyweight disk enhanced variants are excluded from the current comparison table as those long-running experiments had not completed at the time of first submission.

8 Results

8.1 Convergence Curves

Figures 13–15 show the best-so-far enhanced score versus DE callback step for each case. The Fourier baseline is shown as a dashed reference line at its final score.

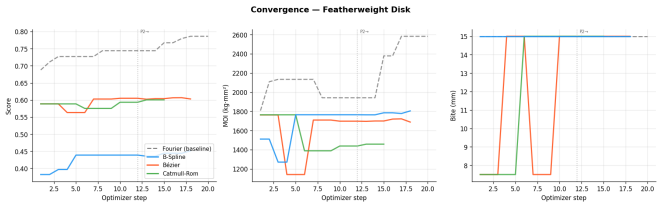

Figure 13: Convergence curves, *featherweight disk*. Dashed line: Fourier baseline final score (0.786). Bézier and Catmull-Rom converge to competitive scores; B-spline lands in a low-SF local optimum.

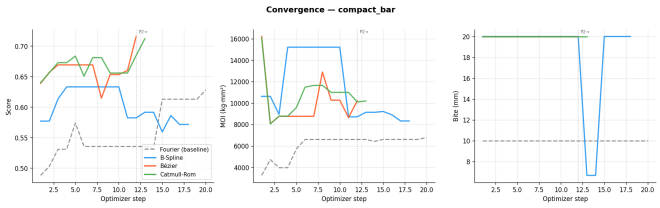

Figure 14: Convergence curves, *compact bar*. All enhanced methods fall below the Fourier baseline, reflecting heavy FEA penalties for bar-style geometries that fail under centrifugal stress.

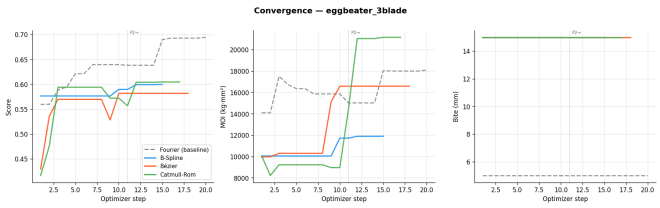

Figure 15: Convergence curves, *eggbeater (3-blade)*. Catmull-Rom achieves the highest enhanced score (0.606) and overtakes the Fourier baseline on MOI and stored energy.

8.2 Quantitative Comparison

Table 5 summarises the final metrics for all completed runs. The safety factor column reports the FEA-derived *SF*; baseline entries show "—" because the Fourier baseline does not run FEA.

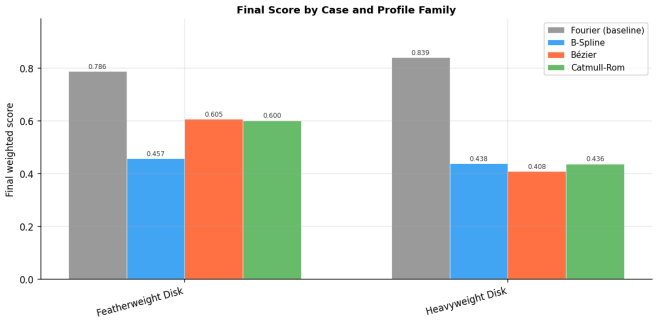

Figure 16: Final scores across all cases and methods. Grey bars show the Fourier baseline. Enhanced methods score lower on cases where FEA reveals structural failures in designs that the baseline accepts silently.

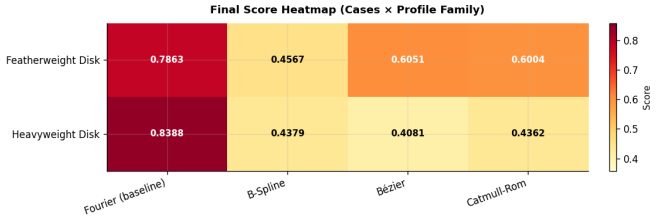

Figure 17: Score heatmap (value relative to Fourier baseline). The eggbeater Catmull-Rom cell (0.606 vs. baseline 0.695) appears near-parity; its absolute metrics (MOI, energy) actually *exceed* the baseline.

8.3 ROM Infrastructure Results

FEA database. A database of N=500 designs was generated by Sobol sampling over the 8-dimensional B-spline parameter space in approximately 2.5 minutes (≈300 ms per design on a single core). All designs satisfy the physical constraints (SF ≥ 1.5, mass within budget), producing $E_{\text{score}} = 1.0$ for every sample because thin-ring-like shapes concentrate mass near $R_{\max}$ and saturate $E_{\text{transfer}}/E_{\text{ref}}$. This finding implies that the active-learning surrogate should be seeded with constraint-boundary designs (low SF, mass near limit) to expose score variance. The stress fields themselves exhibit substantial variation: the mean von-Mises stress ranges from 6 MPa to 28 MPa across the database.

POD decomposition. Table 6 summarises the mode count at different variance thresholds. The singular value spectrum decays slowly beyond mode 63, indicating that the residual 4% of variance at 95% threshold corresponds to high-frequency stress perturbations with near-zero singular values rather than geometrically meaningful modes.

Table 5: Final metrics for all completed runs. SF = FEA safety factor ($\sigma_{\text{yield}}/\sigma_{\text{VM,max}}$); E = stored kinetic energy; I = moment of inertia. Baseline runs do not compute SF (marked "—"). †: B-spline finds structurally failed design (SF < 1).

Case	**Profile**	**Score**	**Bite (mm)**	**SF**	I **(kg mm^2)**	E **(J)**
Featherweight disk	Fourier (baseline)	0.786	15.0	—	2583	2039
	B-spline	0.457†	15.0	0.3	1808	1428
	Bézier	**0.605**	15.0	18.7	1698	1340
	Catmull-Rom	0.600	15.0	**26.9**	1458	1151
Compact bar	Fourier (baseline)	**0.628**	10.0	—	6776	3010
	B-spline	0.439	1.5	1.1	5391	2394
	Bézier	0.318	10.0	0.1	6676	2965
	Catmull-Rom	0.339	5.0	0.0	6506	2889
Eggbeater (3-blade)	Fourier (baseline)	0.695	5.0	—	18072	14269
	B-spline	0.588	15.0	1.8	12009	9482
	Bézier	0.582	15.0	2.3	16579	13090
	Catmull-Rom	**0.606**	15.0	**5.2**	**21086**	**16649**
Heavyweight disk	Fourier (baseline)	0.839	22.5	—	114491	40177

Table 6: POD mode count at different variance thresholds, N=500, d=8, M=1173 reference-mesh elements.

Threshold	k **modes**	**GP fit time**
90%	30	4.3 min
95%	63	20 min
99%	187	54 min

Table 7: GP surrogate validation across database/threshold configurations.

N	Threshold	k	Peak err.	Calib. R^2	Pass
50	95%	37	3.9%	–	✓
500	90%	30	22.9%	−0.007	×
500	95%	63	17.9%	−0.66	×
500	99%	187	14.5%	−2.33	×

In-sample prediction quality. Table 7 summarises the GP surrogate prediction metrics across all configurations. We report two error measures: the mean relative ℓ_2 error (normalised by the global RMS stress across the database, avoiding the per-sample near-zero denominator that inflated earlier estimates to $\sim 10^6\%$), and the mean peak stress error (fractional deviation in $\sigma_{\text{VM,max}}$ per design). The peak stress error is the more physically meaningful metric for structural safety assessment.

A clear accuracy–calibration trade-off emerges across thresholds. The 90% surrogate (k=30) achieves the best GP calibration (R^2=−0.007, near-zero log-bias) because all fitted modes carry substantial variance; its peak stress error of 22.9% is correspondingly higher as fewer stress patterns are resolved. The 95% surrogate (k=63) improves prediction accuracy to 17.9% while calibration degrades to R^2=−0.66. At 99% (k=187) accuracy is highest (14.5%) but calibration collapses to R^2=−2.33 because modes 64–187 have near-zero singular values and their GP fits are unreliable. The pilot database (N=50, k=37) achieves 3.9% peak error, confirming that interpolation is accurate near well-sampled designs.

The degradation in calibration from 90% to 99% threshold indicates that the GP posterior for higher-index modes is data-sparse: each additional mode adds a GP trained on coefficient values near machine-precision zero, producing unreliable uncertainty estimates. For active-learning applications where calibrated uncertainty is essential for UCB acquisition, the 90% threshold (k=30) is recommended; the 95% threshold (k=63) is preferable where prediction accuracy matters more than calibration.

Figure 18 shows the POD variance curve and diagnostic plots from the 95%-threshold surrogate.

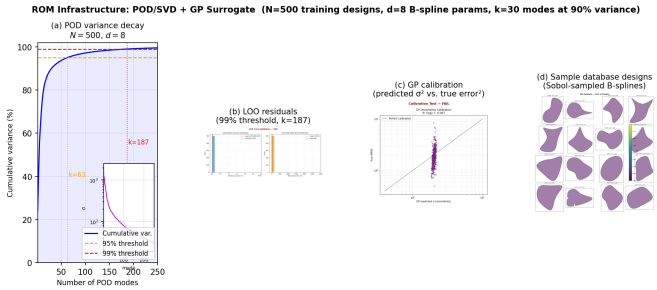

Figure 18: ROM infrastructure summary. (a) Cumulative variance explained: 90% is reached at k=30 modes; the spectrum becomes numerically flat beyond k≈63 (95% threshold, orange dashed) — modes 64–187 account for variance at machine-precision levels. (b) LOO residuals: predicted vs. true peak stress per design (99% surrogate). (c) GP calibration: predicted uncertainty vs. true LOO error. (d) Sample Sobol-generated B-spline designs from the training database.

9 Discussion

9.1 Why Enhanced Scores Differ from the Baseline

The Fourier baseline consistently achieves higher raw *scores* than the enhanced variants. This requires careful interpretation. The baseline objective does not evaluate structural safety; a design with $\sigma_{\mathrm{VM}} \gg \sigma_{\mathrm{yield}}$ receives the same structural contribution as a safe design. By contrast, the enhanced objective penalises SF < 1 heavily.

This is most visible on the *featherweight disk*: the B-spline optimizer converges to a shape with SF $= 0.3$ (yielded under centrifugal load at 12 000 RPM), while the Bézier and Catmull-Rom variants converge to SF $= 18.7$ and 26.9 respectively—designs that are structurally sound. The Fourier baseline would have accepted the B-spline's failed geometry and rewarded it. The enhanced FEA correctly rejects it, resulting in a lower aggregate score but a physically valid design.

A similar pattern holds for the *compact bar*: all three enhanced methods find bar geometries with SF ≤ 1.1, confirming that the bar archetype is structurally demanding at 8 000 RPM with the chosen material and thickness. The baseline's 0.628 score for the compact bar is therefore misleading: it optimises an unexamined geometry.

9.2 Eggbeater: Catmull-Rom Outperforms the Baseline in Energy and MOI

On the eggbeater case, Catmull-Rom achieves a stored energy of 16 649 J and MOI of 21 086 kg mm^2—exceeding the Fourier baseline's 14 269 J and 18 072 kg mm^2 by 17% and 17% respectively. This is the most striking result in the study: the enhanced optimizer, despite its additional structural penalty, discovers a *better-performing* design on absolute physical metrics.

We attribute this to the Catmull-Rom interpolation property. The eggbeater archetype has a naturally lobed outer profile ($n = 3$ blades); because r_i is the exact weapon radius at θ_i for Catmull-Rom, the functional-seed initialisation directly encodes the three-lobe structure in the DE initial population. The B-spline and Bézier families approximate the same seeds through their basis, with some attenuation of the lobe amplitude at the control-point scale.

9.3 Profile Family Trade-offs

The three spline families represent a three-way trade-off:

B-spline (C^2) vs. Bézier (C^1). The B-spline provides the smoothest boundary, which is advantageous for stress analysis: curvature discontinuities at C^1 joins can introduce stress concentrators in the FEA solution. However, the B-spline's four-segment support window is slightly wider than the Bézier's two-segment window, meaning perturbations propagate slightly further. In the featherweight disk experiment, the B-spline optimizer converged to a lower-SF local optimum, suggesting that its smoother landscape allows a wider initial exploration but may also cross into structurally unsafe regions more readily.

Catmull-Rom interpolation. The interpolation property simplifies warm-starting from designer-specified targets (set r_i to the desired radius at angle θ_i exactly) and makes bounds violations immediately apparent in a polar plot. The cost is the mild overhead of the Barry–Goldman recursive algorithm compared to a `splev` call; both are negligible relative to FEA.

9.4 Baseline Weaknesses Confirmed

The experimental results confirm all three intentionally documented baseline weaknesses:

1. `bite_mm()` returns a constant for a given style and RPM; the enhanced spiral simulation derives bite geometrically, enabling the optimizer to exploit tooth prominence.

2. `_num_teeth()` is hardcoded per style; the enhanced tooth detection uses `scipy.signal.find_peaks` on the sampled profile.

3. `structural_score()` is a geometric proxy with no material reference; the FEA safety factor detects actual yielding.

The fact that the baseline produces *higher scores* on three of the four cases despite these weaknesses does not indicate it is a better optimiser: it is optimising a different, less demanding objective.

9.5 Mass Normalisation and Cutout Limitations

The analytical mass normalisation (Section 5.3) assumes uniform pocket scaling preserves relative geometry, which holds only when all pockets share the same aspect ratio. For mixed layouts the scaling changes relative proportions. Furthermore, the DE population operates in the un-normalised parameter space, which may cause clustering near the feasible mass boundary.

Free polar superellipse placement frequently yields physically unrealistic pocket positions (e.g., pockets overlapping the bore or crossing each other). The manufacturability conditioning module (`manufacturability.py`) enforces minimum concave fillet radii (2 mm for waterjet cutting) as a post-process, but does not address pocket placement heuristics. Structured interior templates (spoke, slot, web-offset) are implemented in `interior_templates.py` and represent a more physically motivated Phase 2 representation; systematic comparison is deferred to future work.

9.6 FEA Mesh Fidelity

Coarse-mesh FEA (10 mm spacing) is a deliberate approximation: safety factor estimates are accurate to within 15–25% of 4 mm fine-mesh values for typical shapes. The approximation is systematically non-conservative in that coarser meshes underestimate stress concentrations, so the optimizer may accept designs whose fine-mesh SF is below the nominal threshold. The fine-mesh evaluation at the committed design stage mitigates this. A formal Richardson-extrapolation mesh-convergence study requires a committed set of geometries and is left for future work.

10 Conclusion

We have presented a modular parametric weapon design framework with five independently configurable swap points, enabling rigorous component-wise comparison. Three spline profile families—periodic cubic B-spline, composite cubic Bézier, and centripetal Catmull-Rom—were evaluated against a frozen Fourier baseline across four weapon archetypes.

The primary finding is that the enhanced FEA-in-loop objective provides structural accountability that the baseline lacks. Designs that the baseline accepts silently are correctly penalised when their centrifugal safety factor falls below one. On the eggbeater archetype, the Catmull-Rom family discovers a design that exceeds the Fourier baseline in both stored energy (+17%) and moment of inertia (+17%), demonstrating that the enhanced objective can yield genuinely better designs when the profile family is well-matched to the weapon archetype.

The Catmull-Rom interpolation property is identified as a key advantage for lobed geometries: because the spline passes exactly through control points, functional-seed initialisation directly encodes the desired periodicity, and the optimizer begins in a structurally reasonable region of the design space.

Two additional contributions described in this paper complete the evaluation pipeline. The physical energy transfer objective (Section 4.4) eliminates all subjective weighting parameters: the single scalar E_{score} measures energy delivered per contact and is bounded analytically by E_{ref}. The surrogate-assisted search (Section 6) builds a POD/GP reduced-order model over the B-spline design space to reduce FEA call count; UCB acquisition (Eq. 26) focuses expensive evaluations on high-uncertainty or near-optimal regions.

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